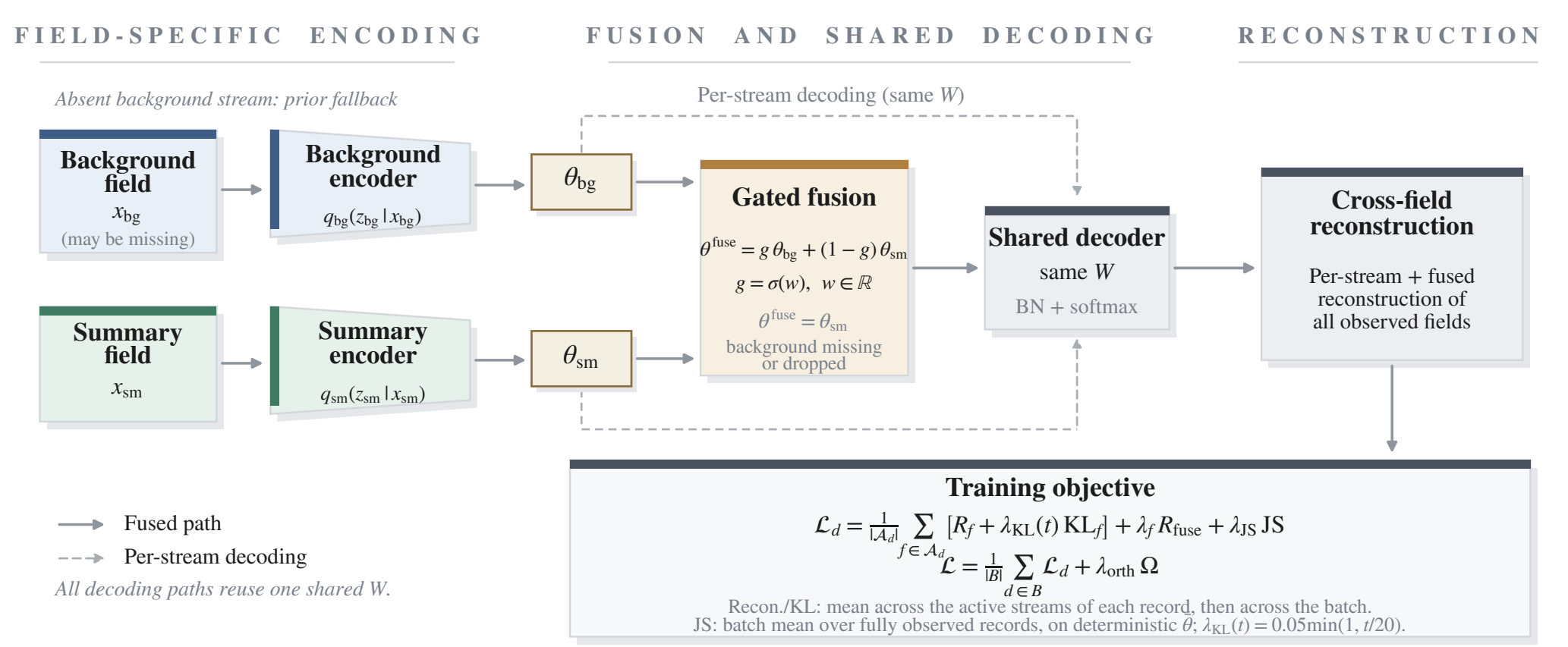
FIELD-SPECIFIC ENCODING
FUSION AND SHARED DECODING
RECONSTRUCTION
Absent background stream: prior fallback
Per-stream decoding (same W)
Background field
x_{bg}
(may be missing)
Background encoder
$q_{bg}(z_{bg} \mid x_{bg})$
θ_{bg}
Summary field
x_{sm}
Summary encoder
$q_{sm}(z_{sm} \mid x_{sm})$
θ_{sm}
Gated fusion
$\theta^{fuse} = g\,\theta_{bg} + (1-g)\,\theta_{sm}$
$g = \sigma(w),\ w \in \mathbb{R}$
$\theta^{fuse} = \theta_{sm}$
background missing or dropped
Shared decoder
same W
BN + softmax
Cross-field reconstruction
Per-stream + fused reconstruction of all observed fields
Fused path
Per-stream decoding
All decoding paths reuse one shared W.
Training objective
$\mathcal{L}_d = \frac{1}{|\mathcal{A}_d|} \sum_{f \in \mathcal{A}_d} [R_f + \lambda_{KL}(t)\,KL_f] + \lambda_f R_{fuse} + \lambda_{JS}\,JS$
$\mathcal{L} = \frac{1}{|B|} \sum_{d \in B} \mathcal{L}_d + \lambda_{orth}\,\Omega$
Recon./KL: mean across the active streams of each record, then across the batch.
JS: batch mean over fully observed records, on deterministic θ; $\lambda_{KL}(t) = 0.05\min(1, t/20)$.